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Web Service Description

For

AZ Payment Gateway

Version 0.3

Prepared by Software Development Department

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Revision History

Name	Reason for Changes	Date	Version
Hassan Elmukashfi	First Creation.	02-05-2024	0.1
Hassan Elmukashfi	Update third case	15-12-2024	0.2
Ahmed SalahEldien	Implement CheckSatus model	28-12-2025	0.3

1.0 Introduction

1.1 Document Purpose

This document is intended to describe the web services provided by AZ Payment Gateway to perform specific transactions. This document also provides brief description of services available and a list of fields required for each service.

1.2 Glossary

JSON	Java Script Object Notation
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2.0 Web Service Description

All transaction requests will be called by external APIs or third party(mobile will be used as example) to AZ payment Gateway.

2.1 Web Service Protocol

AZ payment Gateway web service is based on HTTP POST/GET with MIME type JSON.

All API requests require the following headers:

Header	Value	Required	Description
Content-Type	application/json	Yes	Request content type
language	en or ar ...etc	No	Response language (defaults to en)

2.2 Web Service Connection

AZ payment Gateway uses HTTP connection over secured VPN (firewall) connections.

Ex: http://[url]:[port]/[api]/[api-version]/[service method]

3.0 Web Service Flow

3.1 Request Process

mobile should issue a first request to get all Categories available as start such that the list will only contains as:

- Category Name.
- Category description
- Category ID.
- Category Image.

Response will only contain active categories, at request time any update done in back-end after the request should be inquiry again.

After displaying categories fetched from previous step and User select a category, the mobile will invoke the backend again to get All billers registered to selected Category i.e.

- If User selects Telecom, then backend should return all telecom companies available
- In sub-List User would expect to see either Zain, sudani or MTN.

Every Biller detail will be as:

- Biller ID
- Biller Name
- Biller description
- Biller Image

When the mobile shows the available list of Billers (service providers) such that the user selects a certain provider, another request to list services available within the Biller (service provider) selected and will response as:

- Service ID
- Service Name
- Service description
- Service Image
- isPayment indicator flag.

Response will only contain active services in biller, at request time any update done in back-end after the request should be inquiry again.

When the mobile's user selects a certain service then a final request to shape the last page will be send with the following Parameters:

- Category ID.
- Biller ID.
- Service ID.
- Language

Backend receives the above parameters and then start to prepare a parameter list that matches the paths selected in scenarios and returns data contains:

Parameter specification	Description	Usage
id	Parameter ID	To Identify the parameter when sent in request.
orderNo	Contains a number that's describes its location within the mobile and its page or receipt ... etc ()	A mobile should start building the mobile page/recipe by this order.
name	The name which will be shown to user i.e., Phone, Meter No	This will vary according to the language sent in request either Arabic -English or any specified language.
description	Short sentence that describes the parameter purpose.	could be shown to the customer.
internalKey	This is the key sent to Payment Gateway API in the Json object	It's not unique. Could be used for frontend purposes.
regex	This contains regular expression that the mobile should match the values against it before sending to backend	Mobile and backend uses this regular expression to validate data in a runtime or request since no other validator are available
inputType	Mobile uses this value which will be shared between mobile and backend to specify how this field will be rendered	This value either: • inputText • menu (dropdown list) • checkbox • textArea • radio buttons • Shared(not visible to customer)
option	Used when the 'inputType' value is menu (drop-down list) or radio buttons	This specifies the multiple values of the input. Backend sends this as key-value pairs.
minValue	Minimum value of this input Used only if value is accepting numbers	For amount and ranges If (-1) then it's any.
maxValue	maxValue of an input used if value accepting numbers	For amount and ranges If (-1) then it's any.
length	Specify the length input and used to limit user input	-1 for data doesn't obey this rule such as data with options or textArea.
paramType	Indicates the data type that user will input.	Could be used for UI purposes (e.g keyboard) or data validation.

3.2 Generating Request

In previous steps the mobile user reaches the point when he will submit the request and mobile should construct the request as follow steps:

- Service ID: collected when the user specifies the required service.
- Language: mobile context language.
- Biller ID: biller selected in scenario #2.
- parameters: this will contain a list of request parameters of the service specified above and will be containing data as follow:
 - id: parameter id from the table in 3.1. or from first step response(if it's two step service).
 - key: parameter internalKey from the table in 3.1. or from first step response(if it's two step service).
 - value: either from user input or from first step response(if it's two step service).
- accountFrom: user account(cannot be null or empty if isPayment indicator is 1).
- UserData: user data object contains user id, user account and other information needed. It's optional only needed when user profile is not handled by mobile(third party) side(depends on bank request).

If above step was successful and request data was generated the mobile invoke a route for performing inquiry or payment.

Backend receives request and then perform mapping and generate service specific request, the response value will be one of three(see below section).

3.3 Service's Response

3.3.1 #1

This service Performs both inquiry and payment in one step thus no other process is performed in the mobile and then mobile should only show the receipt. Back end will send data as follow:

- **Response Code:** API indicator whether the process was successful or not or pending(need post-check process see section 3.4). Default value is zero which means successful.
- **Response Message:** message shows either the error and will be in Multilanguage support depending on language in request header.
- **bankReference:** contains the unique reference id for the process.
- **Response_params:** list of json objects contain:
 - Key (represents internalKey as in the table in 3.1)
 - Value
 - Display Name
 - Id. (represents id as in the table in 3.1)
- **Final_status:** this property will be <=0 if the process flow finish else will contains the next service ID that will be sent to API to make the next process.
- **Request_params:** contains the next parameters details as in the table in 3.1 which will be used to construct the next request.
- **isPayment:** indicator related to the service in the next step.

Mobile should check both **response code** and **Final_status** parameters to decide showing receipt or error message.

3.3.2 #2

This service Performs inquiry first as a sperate request then it will be two steps process and Back end will send data with changes in Request_params and final_status parameters:

- **Request_params:** will be having data with structure described in previous section.
- **Final_status:** will contains value >0, which will be the next Service ID in Payment phase.

Mobile should generate a new page that shows data received from process in Response_params plus the

user input from previous step and prompt user to input other data specified in Request_params received in response.

Mobile should always check the parameter **isPayment** whether it's true or false to prompt Account selection for payment.

3.3.3 #3

The mobile captures in response the value **isPayment** is **true** then mobile should prompt user to input Account which payment will be through and generate request as described in 3.2. in addition the parameters list will contain:

- **Previous step user input:** the user input from previous step.
- **Previous step response params:** parameters returned in previous step response in 'response_params' list.
- **Current step user input:** the user input from current step.

Back end performs its process and then return data as described in scenario #2 with changes in:

- **Final_status:** will be ≤ 0 .
- **Request_params:** will be **empty** or **null**.
- **Response_params:** will contains all data to be displayed with its order ID.

3.4 checkStatus

This service allows the client system to check the current status of a payment transaction using its unique transaction ID. It returns the transaction status, description, and details if available.

It returns the transaction status, description, and details if available.

3.4.1 Endpoint

URL: http://[url]:[port]/[api]/[api-version]/checkStatus

3.4.2 Request

3.4.2.1 Request Body

Field	Type	Required	Description
transactionId	string	Yes	The unique transaction identifier to be checked.(returned in 'bankReference' in the payment response)

Example

```
{
  "transactionId": "123333333"
}
```

3.4.3 Response

3.4.3.1 Success Response (HTTP 200)

Field	Type	Description
responseCode	number	CheckStatusReq return code 0 indicates successful status retrieval.
responseMessage	string	Message describing the response status.
trnStatus	number	Original transaction status code. 0 indicates successful status retrieval from the provider. 10 indicates failed status retrieval from the provider.

		20 indicates pending status retrieval from the provider.
trnStatusDisc	string	Description of the original transaction's status.
transactionId	string	The original transaction ID in the system.
checkReqId	string	The internal reference ID for this check request.
trnDetails	array	List of additional original transaction key-value details.

Example

```
Response (success) :
{
  "responseCode": 0,
  "responseMessage": " Successful",
  "trnStatus": 0,
  "trnStatusDisc": "Successfull",
  "transactionId": "851957960024100",
  "checkReqId": "851957960024100",
  "trnDetails": [
    {
      "key": "mobile",
      "value": "0907812682",
      "displayName": "Phone Number ",
      "id": 1
    },
    {
      "key": "responseMessage",
      "value": "success",
      "displayName": "Response Message",
      "id": 3
    }
  ]
}
```

3.4.3.2 Failure Response (HTTP 400 or 200 with Error Code)

Field	Type	Description
responseCode	number	Non-zero value indicates an error.
responseMessage	string	Description of the error.
trnStatus	number	null in case of error.
trnStatusDisc	string	null in case of error.
transactionId	string	Transaction reference.
checkReqId	string	Check request reference.
trnDetails	array	Empty list when an error occurs.

Example

```
{
  "responseCode": 10,
  "responseMessage": "Error while checking transaction status",
  "trnStatus": null,
  "trnStatusDisc": null,
  "transactionId": "851957960024100",
  "checkReqId": "851957960024100",
  "trnDetails": []
}
```

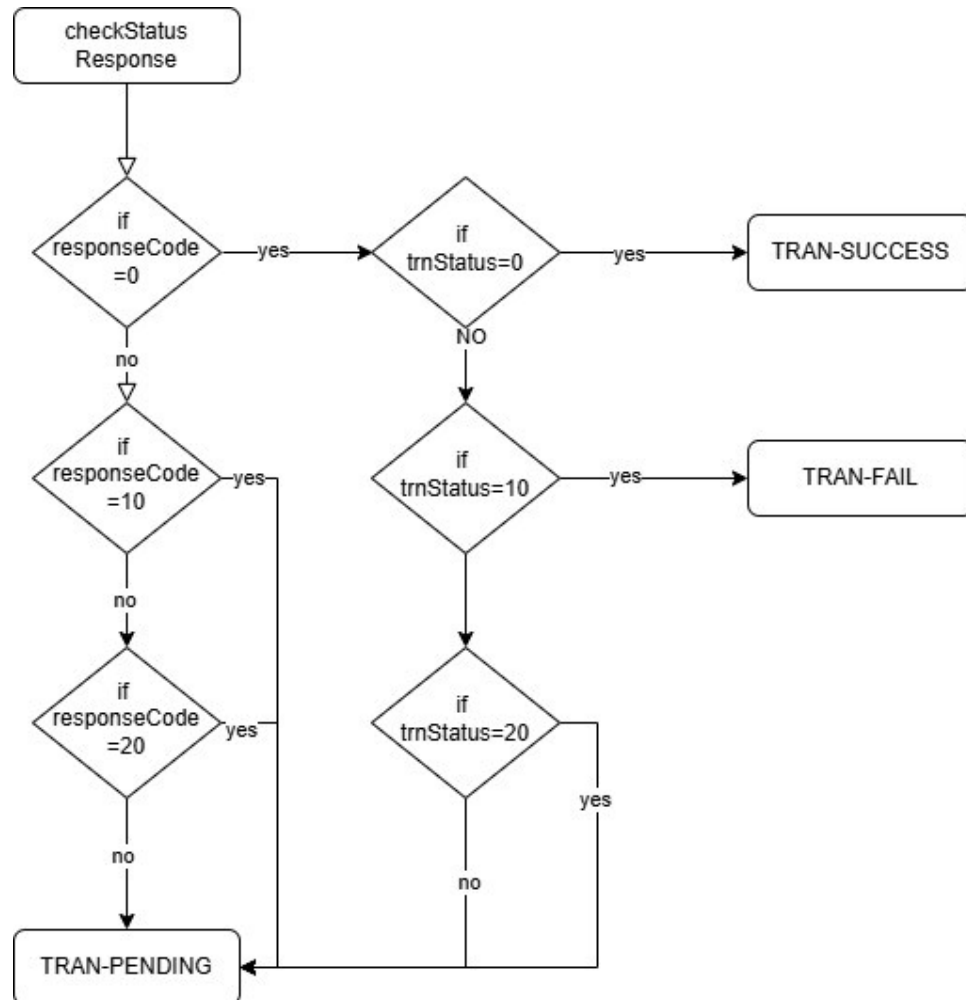
3.4.3.3 Response Codes

Code	Meaning
0	Success – Transaction Status retrieved successfully.
10	indicates failed status retrieval.
20	timeout while getting the transaction status / or indicates the the transaction still pending

Hint: these codes are applied for both `responseCode` and `trnStatus`.

3.4.4 Action FlowChart

This to chart represents how to deal with response codes defined above to a achieve the proper scenario



3.4.5 Notes

- The **transactionId** should be a valid reference previously generated by the payment system.
- Ensure proper error handling for cases where **responseCode** is non-zero.
- The **trnDetails** array may vary depending additional attributes available.